

Project #: _11__

Project Title: ___Digital twin model of 3 motor system fault tolerant system

Category: A / B / C / D / **E**(circle one)

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Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?

The most difficult challenge was deriving an accurate mathematical model for the coupled behavior of three motors while incorporating disturbances and fault conditions. Handling multiple dynamic variables such as torque, angular velocity, and load interactions made the equations complex.

We initially faced confusion in choosing between Newton's method and Lagrange's method. We resolved this by breaking the system into individual motor models using Newton's laws and then combining them systematically. Additionally, we verified our equations step-by-step using simulation tools, which helped us validate and correct errors in the derivation.

Q2: Which Course Outcome (CO1-CO5) was most challenging for your team? Why?

CO3 (Mathematical Modeling and Analysis of Dynamic Systems) was the most challenging. It required translating real-world motor behavior into differential equations and then converting them into state-space form.

The difficulty came from ensuring that the assumptions made during modeling did not oversimplify the real system. Understanding the physical meaning behind each term in the equations also required deep conceptual clarity.

Q3: What would you do differently if you started the project over?

If we started again, we would first create a simplified prototype model before attempting the full multi-motor system. This would help us understand the dynamics more clearly.

We would also begin simulation earlier using tools like MATLAB/Simulink or PyBullet, instead of completing the entire derivation first. Early visualization could have reduced errors and saved time.

Q4: What did you learn from this project that a regular textbook problem could NOT teach you?

This project taught us how theoretical equations connect to real-world systems. Unlike textbook problems, we had to deal with incomplete data, assumptions, and system uncertainties.

We also learned how faults affect system behavior and how self-healing mechanisms (like feedback control or PID correction) can stabilize the system. Most importantly, we understood the importance of validation through simulation rather than relying only on derivations.

Q5: How did you divide work among team members? Did everyone understand the full derivation?

The work was divided into three main parts: mathematical modeling, simulation, and hardware design. Each member initially focused on one area based on their strengths.

However, we conducted regular discussions to ensure everyone understood the complete derivation and system behavior. By the end of the project, all members had a clear understanding of both the theoretical and practical aspects.

Q6: What AI tools did you use? For what specific purpose? What did you NOT use AI for?

We used AI tools like ChatGPT to clarify concepts, assist in deriving equations, and guide us in simulation setup (such as MATLAB/Simulink and PyBullet). It was especially helpful in breaking down complex steps and suggesting structured approaches.

However, we did not rely on AI to directly generate final answers or complete the project independently. All derivations, validations, and understanding were done by the team to ensure conceptual clarity and originality.